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Interleaved incremental association Markov blanket as a potential feature selection method for improving accuracy in near-infrared spectroscopic analysis

Kyeol Chang[a], Junghye Lee[b,**], Chi-Hyuck Jun[b,*], Hoeil Chung[a,**]

[a] Department of Chemistry and Research Institute for Convergence of Basic Sciences, Hanyang University, Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea

[b] Department of Industrial and Management Engineering, Pohang University of Science and Technology (POSTECH), 77 Cheongam-Ro, Pohang 790-784, Republic of Korea

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ABSTRACT

The interleaved Incremental Association Markov Blanket (inter-IAMB) is described herein as a feature selection method for the NIR spectroscopic analysis of several samples (diesel, gasoline, and etchant solutions). Although the Markov blanket (MB) has been proven to be the minimal optimal set of features (variables) that does not change the original target distribution, variables selected by the existing IAMB algorithm could be redundant and/or misleading as the IAMB requires an unnecessarily large amount of learning data to identify the MB. Use of the inter-IAMB interleaving the grow phase with the shrink phase to maintain the size of the MB as small as possible by immediately eliminating invalid candidates could overcome this drawback. In this report, a likelihood-ratio (LR)-based conditional independence test, able to handle spectroscopic data normally comprising a large number of continuous variables in a small number of samples, was uniquely embedded in the inter-IAMB and its utility was evaluated. The variables selected by the inter-IAMB in complexly overlapped and feature-indistinct NIR spectra were used to determine the corresponding sample properties. For comparison, the properties were also determined using the IAMB-selected variables as well as the whole variables. The inter-IAMB was more effective in the selection of variables than the IAMB and thus able to improve the accuracy in the determination of the sample properties, even though a smaller number of variables was used. The proposed LR-embedded inter-IAMB could be a potential feature selection method for vibrational spectroscopic analysis, especially when the obtained spectral features are specificity-deficient and extensively overlapped.

1. Introduction

Near-infrared (NIR) spectroscopy has been widely adopted for the analysis of diverse samples such as agricultural products [1,2], petroleum-driven products [3,4], and polymers [5,6]. Absorption bands in the NIR range are inherently broad due to the nature of the combination and overtone vibrations, and thus, the overall NIR spectral features are severely overlapped and the signature of a target component is barely distinguishable when the sample composition is complex. Therefore, feature selection methods choosing appropriate variables in feature-indistinct spectra have been extensively studied as a potential means to improve the accuracy of measurements. It is generally agreed that a model constructed using only appropriate variables is advantageous for more accurate and robust analysis than one incorporating all the variables, as some of the selected variables may be irrelevant or redundant to the target property, potentially degrading the accuracy of the model [7,8]. Feature selection provides additional benefits, such as enhanced interpretability of multivariate analysis results and reduction of the memory requirements [7,8].

As one of feature selection methods, the usefulness of Markov blanket (MB) feature selection has been demonstrated in the field of machine learning [9,10] especially in bioinformatics such as analysis of genomic microarray data and mass spectra of diverse biological samples [11,12]. MB, originated from a probabilistic graphical model (Bayesian network specifically), is a theoretically proven to be a minimal subset of features that is enough to determine the probability distribution of the target [13]; therefore, it is able to lead to optimal prediction. By incorporating the concept of conditional independence (CI), MB feature

* Corresponding authors.
** Co-corresponding authors.
E-mail addresses: junghye_lee@postech.ac.kr (J. Lee), hoeil@hanyang.ac.kr (H. Chung).

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selection can simply and quickly obtain an optimal feature subset while considering relevance to a target variable and redundancy among features concurrently.

For MB feature selection, an MB discovery algorithm finding the MB of a target variable among features is necessary and CI test of identifying CI between features is basically incorporated. The performance of MB feature selection depends on the MB search strategy and the type of CI test. The most popular MB discovery algorithm is Incremental Association Markov Blanket (IAMB), which first implements the grow phase to include all features that are dependent of a target variable, and then the shrink phase to remove invalid candidates in a separate step. Recently, IAMB has been reported as a potential feature selection method and its utility has been demonstrated for the analysis of diverse kinds of data [14]. However, variables selected by the IAMB could be redundant and/or misleading since the IAMB requires an unnecessarily large amount of learning data to identify the MB [15–17]. Alternatively, interleaved IAMB (inter-IAMB) [16,17] which interleaves the grow phase with the shrink phase to maintain the size of MB as small as possible by immediately eliminating invalid candidates is potentially more effective in feature selection.

As mentioned above, although the inter-IAMB could be a potential feature selection method, its acquaintance and application for multivariate spectroscopic analysis are still at an introductory stage. Therefore, this publication attempts to present the inter-IAMB as an effective feature selection tool for the analysis of vibrational spectroscopic data. Usually, a sample spectrum consists of a large number of variables, while the number of employed samples in a dataset is small. Therefore, to handle this kind of spectroscopic data, a likelihood-ratio (LR) was embedded in this work as a CI test in the inter-IAMB rather than the conventionally employed Student's *t*-test or Fisher's *z*-test [18–20]. The details of the inter-IAMB and LR-based CI test are described in the following section. For the evaluation, NIR spectral datasets were separately collected from diesel, gasoline, and etchant solution samples. Diesel and gasoline samples, which are complex mixtures composed of a large number of hydrocarbons with different structures and chain lengths, were used to evaluate the performance of the inter-IAMB since the NIR spectral features of these mixtures are severely overlapped. Meanwhile, the etchant solutions employed here contained four different inorganic acids, which do not directly absorb NIR radiation. Therefore, the perturbed water bands of these components are the only available source for analysis, while their NIR spectral features are considerably indistinct and deficient of component-specificity. Under these situations, the selection of appropriate variables and use of the chosen variables for the measurement of the corresponding properties could improve the accuracy of their analysis. Thus, the variables selected by inter-IAMB in the NIR spectra were used to determine the sample properties (see Table 1) using either multiple linear regression (MLR) or partial least squares (PLS). In parallel, the properties were also determined using the IAMB-selected variables and all the variables, and the resulting accuracies were compared to those obtained using the inter-IAMB-selected variables.

2. Interleaved incremental association Markov blanket (inter-IAMB)

The process for the identification of variables by the MB is similar to that of a filter method [21], and the IAMB algorithm is the most commonly adopted one [16]. As shown in Fig. 1, the IAMB consists of two separate steps [22]: finding the set of all MB candidates that describe a target property (grow phase) and discarding invalid candidates in the MB set (shrink phase). In the grow phase, the variables determined to be conditionally dependent on the target property are added to the MB through a CI test based on Fisher's *z*-test. All variables are initially ranked in order of strength of their dependency on the target property. The strength of the dependency is estimated by the *p* value of the *z*-test. The variable with the smallest *p* value is selected as the first candidate and included in MB, if it is determined to be dependent on the target property by hypothesis testing. Next, variables other than the selected one are evaluated in the same fashion and a second variable is determined. This procedure is repeated until no further variables are added to the MB. In the shrink phase, among all the variables selected in the grow phase, the variables that are conditionally independent of the target property are removed. Specifically, each variable is sequentially considered in the same order as in the previous grow phase and then, whether a variable is dependent on the target variable or not is determined when the rest of the features in the MB are conditioned. When a variable is found to be conditionally independent, it is regarded as a false positive of the MB and is thus removed from the MB. This process continues until all the selected variables in the MB are individually screened.

Table 1
Description of the measured sample properties, the corresponding ranges of variation, the number of samples in each dataset, and the number of variables in the corresponding spectrum.

Sample	Number of samples	Number of variables	Property	Range
Diesel	87	416	Cetane index	54.7 − 59.5
			T 50%	261.1 − 282.8 °C
Gasoline	99	416	RON	91.1 − 100.3
			T 50%	66.1 − 105.8 °C
Etchant solution	80	675	HCl concentration	0.1 − 2.0 M
			HNO_3 concentration	0.1 − 2.0 M
			H_2SO_4 concentration	0.1 − 2.0 M
			H_3PO_4 concentration	0.1 − 2.0 M

However, the variables selected by IAMB could be redundant and/or misleading since the IAMB requires an unnecessarily large amount of learning data to identify the MB [15–17]. This tendency can negatively influence accuracy of analysis. As shown in Fig. 1, the IAMB adds all candidate variables and then removes invalid candidates in a separate step. In contrast, in the case of the inter-IAMB (see the pseudo-code in the Supplementary Data), when a candidate variable is added to the estimated set of the MB (corresponding to lines 1–5 in the pseudo-code), variables becoming invalid by the addition of the candidate variable are immediately eliminated (lines 6–10). More specifically, inter-IAMB interleaves grow phase with shrink phase to maintain the MB size as small as possible during all the steps of the algorithm's execution. Therefore, the IAMB is data-inefficient because it may require an unnecessarily large amount of learning data to identify the MB, as stated earlier, while the inter-IAMB is more efficient by using fewer data and has the ability to avoid false positives during the course of MB detection.

Usually, for CI testing in the MB discovery algorithm, a G^2 test or a χ^2 test (count-based CI tests) [15] using contingency tables is usually implemented to handle categorical features, while a Fisher's *z*-test or a Student's *t*-test (model-based CI tests) [15] employing linear regressions is used to find continuous features. Although MB feature selection can be applied for classification and regression employing different CI tests, it can only deal with data of the same type as that of the target. For example, it can be used for classification with categorical features and for regression with continuous features. When a dataset is composed of a small number of samples with a large number of variables, such as vibrational spectroscopic data, the use of a likelihood-ratio (LR) is known to be more effective for the CI test, and was thereby employed in this study. The test compares the goodness of fit of the null and the alternative models, where the null model is a special case of alternative model (i.e., nested models). More specifically, it is based on the LR Λ,

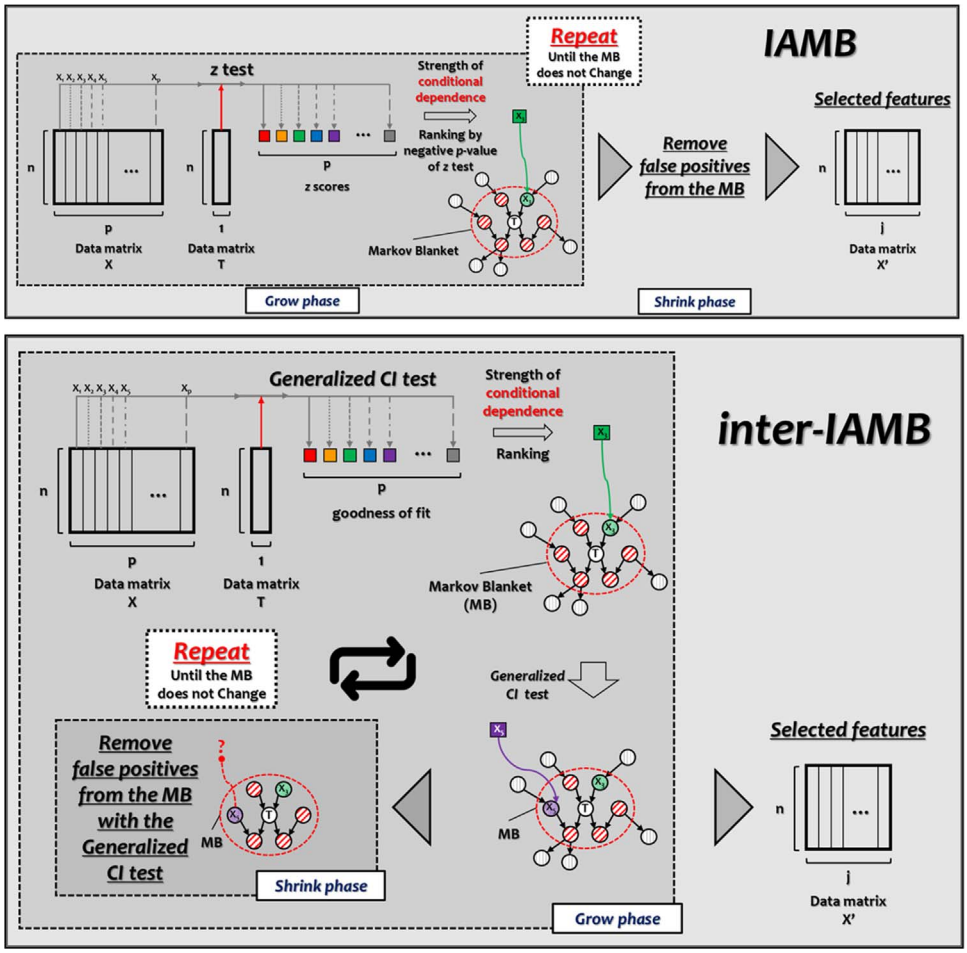

Fig. 1. Graphical presentation of the procedure for variable selection in the IAMB (top) and inter-IAMB (bottom) methods.

which expresses how many times certain data are more likely under one model than the other with the test statistic of –2logΛ. The test statistic approximately follows a χ^2 distribution with degrees of freedom (df) equal to the difference between the dfs of the alternative model and the null model [23].

Based on this fact, we constructed a problem-specific model, from which the likelihood could be obtained, and used the LR test to determine the CI. In this study, both the target and features were continuous and thus multiple linear regression (MLR) models were used. For example, let Y be a continuous target, X be a candidate feature with a continuous value, and Z be a given continuous feature. Then, we can construct the following MLR model to obtain the LR and perform the LR test:

$$Y = \alpha + \beta X + \gamma Z$$

which determines the CI between X and Y given Z under the null hypothesis $H_0 : \beta = 0$ and df = 1. If the null hypothesis is true, then X and Y are conditionally independent for a given Z. The given feature Z can be extended to a multivariate case Z, and the test is performed in the same way. Here, each instance of Y, Y_i, $i = 1, \ldots, n$ is assumed to follow a multivariate normal distribution on X_i and Z_i with a constant variance of σ^2. Therefore, the conditional probability density function of Y_i is

$$f_Y(Y_i, X_i) = (2\pi\sigma_0^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\frac{(Y_i - \alpha - \beta X_i - \gamma Z_i)^2}{\sigma_0^2}\right),$$

and the likelihood function is

$$L(\beta, \gamma, \sigma_0^2; Y, X) = (2\pi\sigma_0^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(Y_i - \alpha - \beta X_i - \gamma Z_i)^2\right).$$

3. Experimental section

3.1. NIR sample datasets and determination of sample properties

Diesel and gasoline samples (87 and 99 samples, respectively) were acquired from a petroleum company in Korea. Etchant solutions (80 samples) containing HCl, HNO_3, H_2SO_4, and H_3PO_4 were prepared by mixing the individual acids in water at concentrations ranging from 0.1 to 2.0 M (increment intervals of 0.1 M). The NIR spectra of all the samples were collected using an ABB MB-3600 FT-NIR spectrometer (Quebec City, Quebec, Canada) equipped with a tungsten-halogen source and an InGaAs detector. Air was used as the background to obtain the absorbance spectra. The pathlengths used for the spectral acquisition of diesel/gasoline and etchant solution samples were 0.5 mm and 2.0 mm, respectively. Table 1 shows the sample properties to measure and the corresponding ranges of variation. All spectral preprocessing and multivariate analyses, including baseline correction, normalization, feature selection (IAMB, inter-IAMB), MLR, and PLS, were performed using MATLAB R2016a (Math Works Inc., MA, USA).

4. Results and discussion

4.1. Evaluation of the inter-IAMB for the determination of diesel and gasoline properties

Diesel is a highly complex mixture of hydrocarbons typically containing between 8 and 21 carbon atoms per molecule. Fig. 2(a) shows the normalized NIR spectra of the diesel samples in the 4800–4000 cm^{-1} range. For the normalization, the baselines of the raw NIR spectra were linearly offset to zero at 4800 cm^{-1} and each

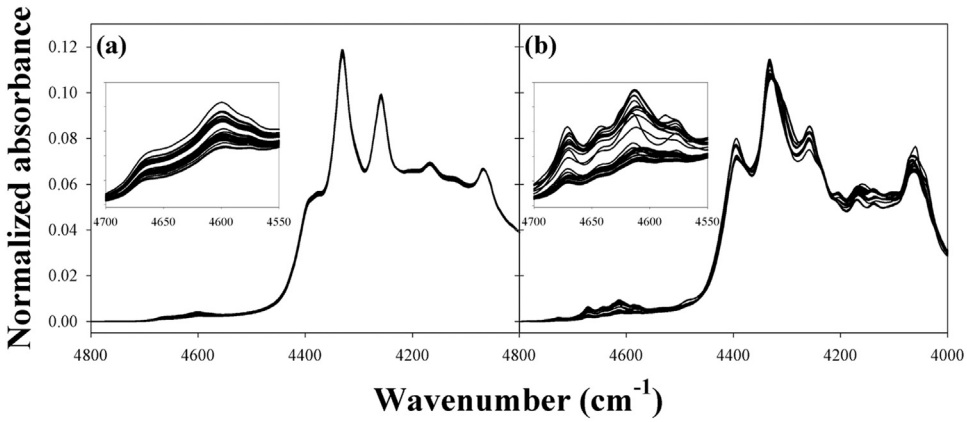

Fig. 2. Normalized NIR spectra of all the diesel (a) and gasoline (b) samples in the 4800–4000 cm^{-1} range. The insets show the 4700–4550 cm^{-1} range.

baseline-corrected spectrum was divided by the corresponding peak area under the 4800–4000 cm^{-1} range. The insets (4700–4550 cm^{-1} range) highlight the combination bands originated solely from aromatic hydrocarbons. The combination bands of various hydrocarbons, such as paraffin, naphthene, and aromatic components, appear in the 4500–4000 cm^{-1} range. The NIR spectral features of the diesel samples are the result of the extensive overlapping of the bands of the individual components; therefore, the selection of appropriate variables before quantitative multivariate analysis would be potentially beneficial. Initially, PLS was employ to determine the cetane index and T50% (temperature at which 50% of a sample is distillated; alternatively, 50% recovery of the distillate) using the whole spectral range. From the whole dataset, 90% of the samples were randomly selected as a calibration set to build the model and the remaining 10% of the samples were used for the validation set. This procedure was repeated 1000 times for cross validation, and finally, the standard error of cross validation (SECV) was obtained. Next, variables were selected using both the IAMB and inter-IAMB methods, and the chosen variables were separately used for MLR to determine the properties of the samples.

Table 2 shows the SECVs resulting from the three different methods described above. The numbers in parentheses indicate the number of factors used in the PLS models. As shown, IAMB or inter-IAMB selected variables were used for the MLR, and the errors in the determination of both cetane index and T50% decreased in comparison with those using PLS with whole variables. The use of inter-IAMB selected variables afforded slightly better accuracy compared to that obtained from IAMB selected variables. Because the number of variables selected by the IAMB and inter-IAMB was too small, MLR was unavoidably chosen to determine the properties of the samples. As shown, The uses of only 5 and 12 variables selected by the inter-IAMB (out of the total 416) were effective to achieve better accuracy in the determination of the cetane index and T50% values, respectively.

Fig. 3(a) and (b) show the location of the IAMB and inter-IAMB selected variables on the diesel spectra for the determination of the cetane index and T50%, respectively. For the measurement of the cetane index, the location of the selected variables is quite similar in both cases, and so their accuracy is nearly similar with each other. Meanwhile, for the measurement of T50%, the location of the inter-IAMB selected variables differs considerably from that of the IAMB selected variables, and thus the improvement in accuracy is more noticeable using the inter-IAMB.

Fig. 2(b) shows the normalized NIR spectra of all the gasoline samples in the 4800–4000 cm^{-1} range. Compared to the diesel spectra, the dissimilarity in the gasoline samples is greater, indicating larger compositional variation among the samples. As it is known, gasoline is a blended product of diverse refinery and petrochemical feedstocks and thus, the resulting NIR spectral features are also complexly overlapped. Using the spectra, the determination of the research octane number (RON, the most important parameter for the assessment of gasoline properties) and T50% was accomplished in the same fashion and the subsequent results were analyzed, as shown in Table 2.

In the case of the RON, the SECV substantially decreased to 0.15 using MLR with the inter-IAMB selected variables, and the use of only nine variables was effective to follow the variation in the RON of the samples. For the measurement of T50%, the use of either IAMB or inter-IAMB selected variables led to improved accuracy compared to that from PLS employing the whole variables. In contrast, the SECV resulted from the use of inter-IAMB selected variables was slightly inferior to that in the case of IAMB selected variables. Although only six variables selected by inter-IAMB were used for the MLR analysis, the resulting accuracy was comparable. It must be noted that the number of IAMB selected variables, 16 variables, would be too large to use for MLR. Thus, the probability of over-fitting would be considerable in the MLR model using the IAMB selected variables.

Fig. 3(c) and (d) show the location of the IAMB and inter-IAMB selected variables on the gasoline spectra for the determination of the

Table 2
SECVs for the determination of the sample properties using all the variables, IAMB-selected variables, and inter-IAMB-selected variables. MLR was adopted in the determination of the diesel/gasoline properties using the IAMB-selected and inter-IAMB-selected variables; in the other cases, PLS was used. The numbers in parentheses indicate the number of factors used for PLS.

Sample	Property	All variables		IAMB selected variables		inter-IAMB selected variables	
		Number of variables	SECV	Number of variables	SECV	Number of variables	SECV
Diesel	Cetane index	416	0.45 (5)	4	0.39	5	0.38
	T 50% (°C)	416	0.96 (9)	14	0.67	12	0.52
Gasoline	RON	416	0.35 (13)	14	0.30	9	0.15
	T 50% (°C)	416	2.78 (4)	16	1.86	6	1.96
Etchant solution	HCl (M)	675	0.040 (10)	76	0.032 (10)	34	0.028 (9)
	HNO_3 (M)	675	0.056 (10)	76	0.047 (10)	36	0.042 (9)
	H_2SO_4 (M)	675	0.048 (10)	137	0.048 (10)	31	0.039 (9)
	H_3PO_4 (M)	675	0.051 (10)	162	0.050 (10)	31	0.040 (9)

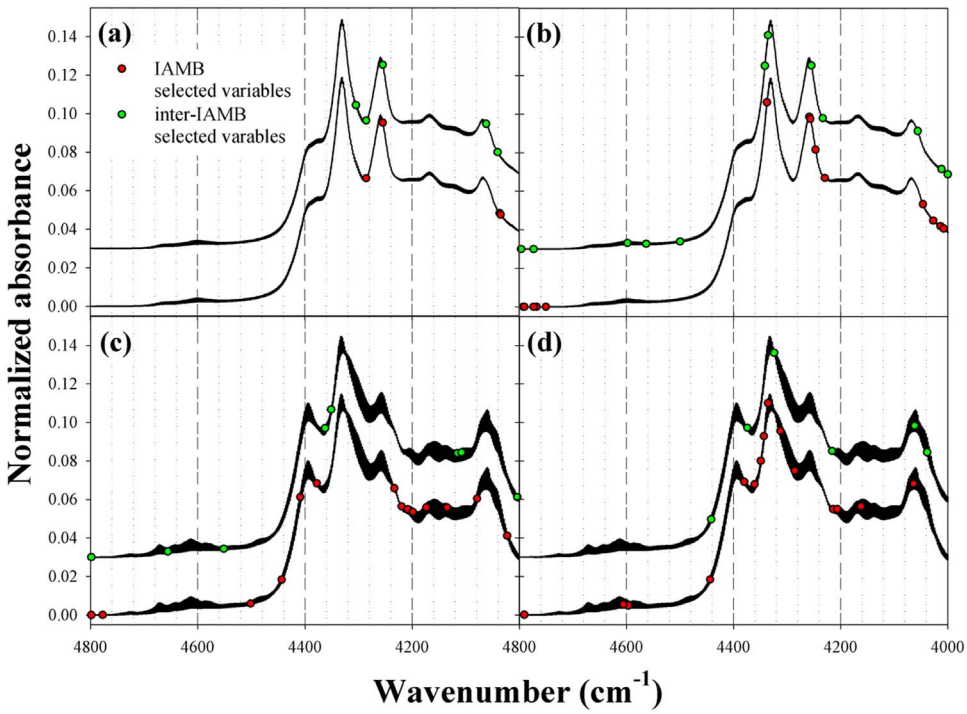

Fig. 3. Location of the IAMB and inter-IAMB selected variables on the diesel spectra for the determination of the cetane index (a) and T50% (b), and on the gasoline spectra for the determination of the RON (c) and T50% (d).

RON and T50%, respectively. For the measurement of the RON, the location of the inter-IAMB selected variables differed considerably from that of the IAMB selected variables. This suggests that the information provided by the inter-IAMB selected variables is certainly more descriptive of the changes in the RON, as supported by the significant decrease of the SECV. On the other hand, for the measurement of T50%, the location of inter-IAMB selected variables was close to that of the IAMB selected variables.

4.2. Evaluation of the inter-IAMB for the compositional analysis of etchant solutions

Wafer etchant solutions are composed of several inorganic acids and the determination of their concentration is required to achieve the desired wafer cleaning performance. Generally, inorganic acids do not directly absorb NIR radiation, and thus the perturbed water bands generated from dissociated hydronium ions (H_3O^+) and the corresponding anion are the main sources for quantitative analysis. Therefore, the NIR bands obtained from inorganic acids are not clearly specific to the measured components, since the water bands are very broad and the interactions of ionized species with water molecules are lack of structure-specificity, as observable in the normalized NIR spectra (9000–7700 cm^{-1}) of 0.5 M HCl, HNO_3, H_2SO_4, and H_3PO_4 samples in Fig. 4. The spectra have been arbitrarily offset for comparison of their spectral features. For the normalization, the baselines of the raw spectra were vertically corrected at 9000 cm^{-1} and the baseline-corrected spectra were then divided by the corresponding integrated areas in the 9000–7700 cm^{-1} range. The broad bands in this range correspond to overtones of the fundamental –OH vibrations [24,25]. It is obvious that changes in the concentration of each acid result in changes in the shape of the water band due to the interactions of dissociated species with water molecules and the displacement of water molecules. However, at a glance, the spectral features of the inorganic acids are quite similar and only very minute component-specific characteristics can be discerned.

Next, mixture samples of these inorganic acids were prepared and their NIR spectra were used to determine the concentration of each inorganic acid. The concentration of each component was varied randomly in the 0.1–2.0 M range to avoid any considerable correlations between the individual concentration variations. Table 2 shows the SECVs resulting from the determination of the concentration of each inorganic acid. One very noticeable observation is that the number of variables selected by either the IAMB or inter-IAMB is significantly larger compared to those in the analysis of diesel and gasoline samples. As mentioned earlier, the NIR spectral features of each inorganic acid are not distinct, and a larger number of variables is required to differentiate the mixed spectral features of the individual components. Therefore, PLS rather than MLR was used to determine the concentration of the components when IAMB or inter-IAMB selected variables were employed. The numbers in parentheses indicate the number of factors used in the PLS models.

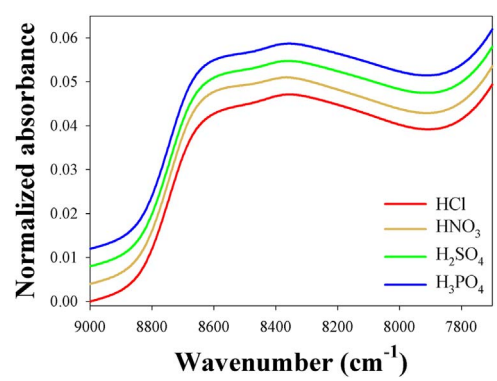

Fig. 4. Normalized NIR spectra (9000–7700 cm^{-1} range) of the 0.5 M HCl, HNO_3, H_2SO_4, and H_3PO_4 samples. The spectra have been arbitrarily offset for clarity.

In all cases, the accuracy in the concentration determination improved when the inter-IAMB selected variables were used for PLS, although the number of employed variables was much smaller (31–36 variables out of the total 675 variables). When the numbers of selected variables by the IAMB and inter-IAMB were compared, the inter-IAMB chose a much smaller number of variables. Nonetheless, such fewer inter-IAMB selected variables were still more descriptive to track the concentration changes of each inorganic acid. Again, this result demonstrates the utility of the inter-IAMB in feature selection.

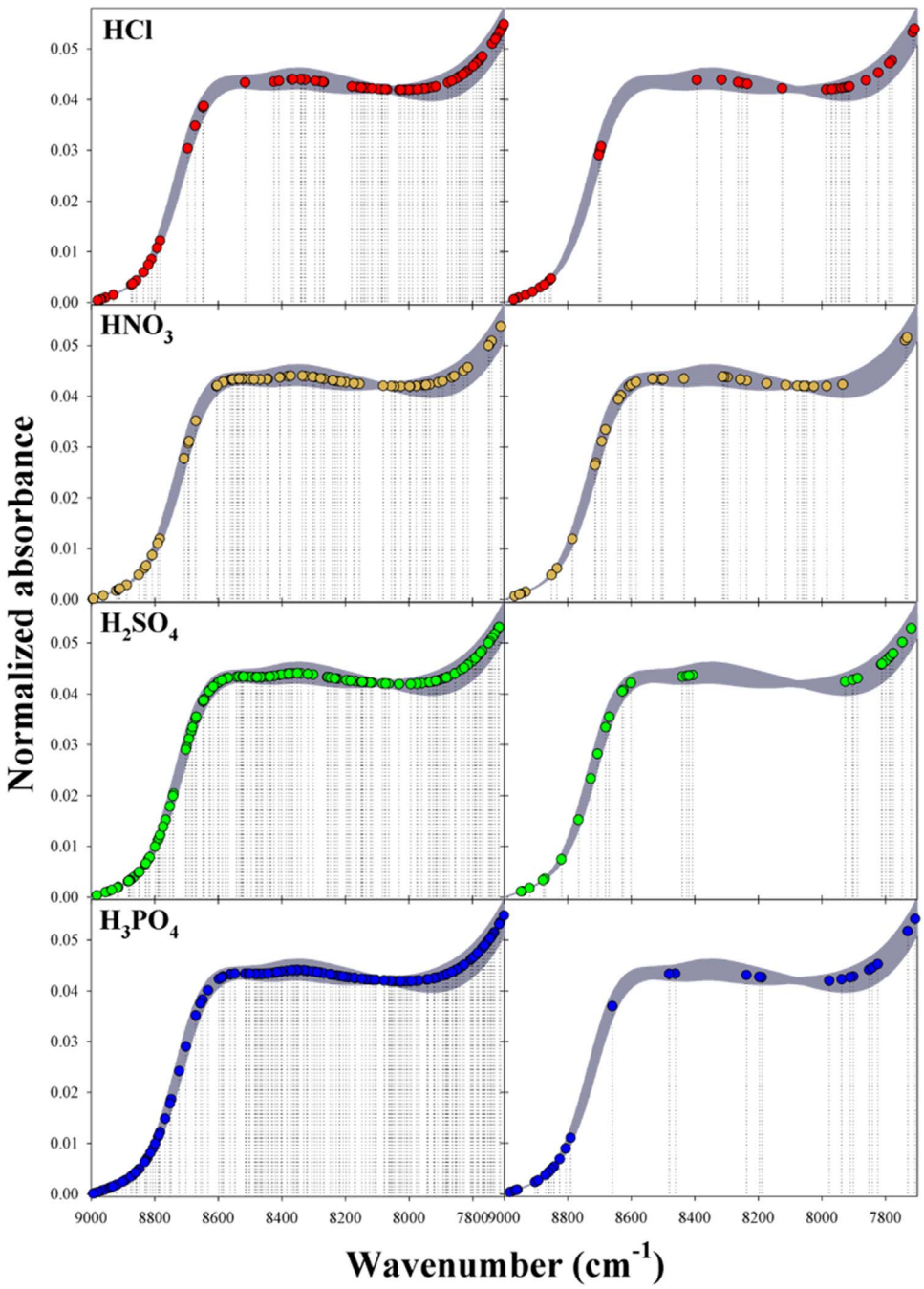

Fig. 5. Location of the variables selected by the IAMB (left column) and inter-IAMB (right column) on the spectra of the mixtures for the determination of the HCl (first row), HNO_3 (second row), H_2SO_4 (third row), and H_3PO_4 (fourth row) concentration.

Fig. 5 shows the location of the selected variables by the IAMB (left column) and inter-IAMB (right column) in the spectra of the mixtures. In the case of the IAMB, the selected variables are rather evenly distributed over the whole spectral range for all four components; therefore, the component-specificity is somewhat questionable. In contrast, the inter-IAMB selected variables are more localized within certain ranges. It can thus be concluded that the inter-IAMB is able to find variables that provide more component-specific information, as evidenced by the improvement in accuracy of the measurements.

5. Conclusions

In this study, the potential of the inter-IAMB as an improved feature selection method has been demonstrated for the analysis of samples with complexly overlapped and feature-indistinct NIR spectra (diesel, gasoline, and etchant solution samples). The inter-IAMB embedded with a generalized CI test based on LR was more effective for the selection of variables than the IAMB and thus able to improve the accuracy in the determination of the properties (or concentrations) of the employed samples, even when using a smaller number of variables. Although we expect that the use of such inter-IAMB may not always lead to the improvement of accuracy in spectroscopic analysis due to the widely different characteristics of target samples, the proposed method could be highly valuable when the obtained spectral features of the samples are specificity-deficient and extensively overlapped. Future research will be devoted to evaluate this methodology for the analysis of other vibrational spectroscopic data with different characteristics, such as those acquired using IR and Raman spectroscopy.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.talanta.2017.09.039.

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